Metal displacement

Answers

Growing metal crystals

1. A shiny copper wire was suspended in a test tube containing a solution of silver nitrate. Over time, shiny silvery coloured crystals grew on the surface of the copper wire. The experiment was repeated with a shiny magnesium wire in place of the copper wire. Once again silvery crystals grew on the wire surface. However, the crystals grew at a faster rate.

(a) Identify the silvery crystals that formed in each experiment.

Silver forms in both experiments.

(b) Write word equations for each reaction.

Copper + silver nitrate → silver + copper nitrate

Magnesium + silver nitrate → silver + magnesium nitrate

(c) Explain why the crystals grew faster in the second experiment.

Magnesium is a more active metal than copper and reacts more rapidly.

The activity series of metals

2. The following table shows the activity series of metals.

Li	K	Na	Ca	Mg	Al	Mn	Cr	Zn	Fe	Ni	Sn	Pb	H	Cu	Hg	Ag	Au	Pt

The photo shows zinc reacting with hydrochloric acid.

Photo: Public domain

(a) Identify the gas released in the reaction.

Hydrogen

(b) Write a word equation for the reaction.

Zinc + hydrochloric acid → hydrogen + zinc chloride

(c) Explain why this reaction is an example of a displacement reaction.

The zinc displaces hydrogen from the solution.

(d) Use the activity series to identify:

(i) metals that will not react with hydrochloric acid

Cu; Hg; Ag; Au; Pt

(ii) the metal that will react most rapidly with the acid solution.

Li (lithium)